

AFREEN AURSHI

Machine Learning & Product Analytics

Lucknow, India | +91 95554 61784 | arshiafreen090@gmail.com
GitHub | LinkedIn | Portfolio

EDUCATION

University of Lucknow, Faculty of Engineering and Technology
B.Tech. in Computer Science and Engineering

2024 – 2028
CGPA: 8.0/10

TECHNICAL SKILLS

Languages: Python, C++, SQL

Machine Learning: scikit-learn, XGBoost, SHAP, Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Time-Series Forecasting, Recursive CV

Data: NumPy, pandas, Matplotlib, Seaborn, Statistical Analysis, Exploratory Data Analysis

Tools: Git, GitHub, Jupyter Notebook, Kaggle, Google Colab, Streamlit, VS Code

Other: Geospatial Analysis, Haversine Distance, BallTree, Data Visualization, H3

MACHINE LEARNING PROJECTS

Delivery ETA Prediction & Dark-Store Analysis | XGBoost, Geospatial ML

GitHub | Live App

- Built an ETA regression workflow on **42,508 delivery records**, engineering Haversine distance features with leakage-safe train-only preprocessing.
- Tuned XGBoost to **MAE 3.01 min, RMSE 3.76 min, and R^2 0.840**; added SHAP explanations and deployed an interactive Streamlit application.
- Analyzed dark-store proximity using **BallTree/Haversine distance** across 287,065 customer-store pairs to evaluate quick-commerce delivery reach.
- Evaluated candidate dark stores by customer proximity and geographic coverage to support **nearest-store fulfillment and last-mile delivery analysis**.

Meal Demand Forecasting | XGBoost, Time-Series Validation

GitHub | Live App

- Modeled **456,548 demand records** spanning 145 weeks, 77 fulfillment centers, and 51 meals; engineered lag, rolling-history, center-meal, promotion, and price features.
- Designed **leakage-safe expanding-window validation** with recursive week-by-week predictions over a 10-week horizon and benchmarked against a latest-value baseline.
- Reduced recursive CV RMSLE from **0.825 to 0.515** with XGBoost; compared ETS, SARIMA, and Prophet and analyzed errors across demand segments.

Heart Disease Prediction | Classification, Streamlit

GitHub | Live App

- Compared five classification models on **918 records**, evaluating model performance using accuracy and F1-score.
- Logistic Regression achieved **90% accuracy and 89% F1**; packaged the trained model into an interactive Streamlit prediction application.

EXPERIENCE & LEADERSHIP

AWS Cloud Club, University of Lucknow | Tech Team Member

Oct 2025 – Present

- Supported a hands-on cloud learning workshop and coordinated across design and technical teams; previously contributed event materials as a Design Team Member.

Connecttly | UX Design & Content Intern

Feb 2026 – Mar 2026

- Designed website UI assets and product-update content in Figma while maintaining consistent layouts and iterating based on team feedback.

ADDITIONAL

Interests: Machine Learning, Data Science, Forecasting, Geospatial Analytics, Product Analytics, Quick-Commerce Operations, AI/ML Systems